

JudgeSense: A Benchmark for Prompt Sensitivity in LLM-as-a-Judge Systems

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May 11, 2026

Abstract

Large language models are widely adopted as automated evaluation judges, yet the stability of their verdicts under semantically equivalent prompt rephrasings remains largely unexamined. We conduct a systematic empirical study of prompt-induced decision instability across multiple evaluation tasks and judge architectures. To facilitate this analysis, we release JUDGESENSE, a benchmark comprising hand-validated prompt-paraphrase pairs spanning factuality, coherence, relevance, and preference, drawn from established NLP benchmarks and accompanied by comprehensive decision logs. The benchmark enables the measurement of judge stability across equivalent prompts, allowing researchers to assess whether stability correlates with model scale or instruction-tuning, and to identify which tasks are most sensitive to prompt wording. Our evaluation reveals that coherence remains the primary task for distinguishing judge behavior, while factuality judgments demonstrate high stability under standard conditions. Pairwise evaluation tasks consistently exhibit position bias. Crucially, we find that model scale is not a reliable proxy for consistency; notably, as an interesting result in our analysis, the largest and newest models are not the most consistent. Code and dataset available at <https://github.com/rohithreddybc/judgeSense> and <https://huggingface.co/datasets/Rohithreddybc/judgesense-benchmark>.

1 Introduction

Consider a familiar setting. A practitioner wants to compare two summarization systems. They use GPT-4o as a judge, prompting it with: “*Rate the coherence of this summary from 1 to 5.*” The next day, a teammate runs the same evaluation with what they consider an equivalent prompt: “*On a scale from 1 to 5, how coherent is this summary?*” The two prompts

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ask the same question. The judge is the same model at temperature zero. The summary being scored is the same. Yet on a non-trivial fraction of items, the score changes. In our data, GPT-4o flips its 1–5 coherence rating of the same summary on 8.5% of pairs under this kind of rephrasing. Smaller, less instruction-tuned judges flip far more often: gemini-2.5-flash flips on 61.3% of the same coherence pairs.

This kind of variation undermines a basic assumption of LLM-as-a-judge evaluation: that the prompt template is an implementation detail, not a confounder. If two researchers pick slightly different templates and get systematically different rankings of the same models, they will draw different conclusions from the same underlying data. The judge stops being a measurement tool and starts being a noise source.

Prompt sensitivity in LLMs as task-solvers has been studied extensively [15–17, 24]. The judge case is different. A judge does not generate an answer; it returns a label, often from a small finite set. Its sensitivity is therefore not about output fluency or content quality, but about decision stability across paraphrases. Three recent lines of work touch on this. Thakur et al. [19] note prompt complexity as a vulnerability among 13 judge models, but treat it as one of several biases rather than as a primary axis of measurement. Arabzadeh and Clarke [1] examine prompt sensitivity in LLM-based relevance judgment, but on a single task with three judges. Shi et al. [18] study a related phenomenon (position bias in pairwise judges) across 12 models and 100,000 evaluation instances, but their concern is option ordering rather than instruction phrasing. None of these works conducts a systematic study of prompt-paraphrase decision stability across multiple evaluation tasks with a fixed, reusable benchmark and a uniform reporting convention.

We operationalize this stability as a simple, reportable score. The Judge Sensitivity Score (JSS) is defined as

$$\text{JSS}(j, t) = \frac{1}{|P|} \sum_{i=1}^{|P|} \delta(j(p_i), j(p'_i)), \quad (1)$$

where j is a judge model, t is an evaluation task, P_t is a task-specific set of *prompt-paraphrase pairs* (p_i, p'_i) , that is, evaluation instructions sharing the same evaluative intent on the same item but differing only in surface phrasing, and the indicator function $\delta(a, b) = 1$ if $a = b$ and 0 otherwise. JSS lies in $[0, 1]$. A value of 1.0 means the judge gave identical decisions on every paraphrase pair; a value below 0.8 means the judge’s verdict can be flipped on more than one in five items by rephrasing alone. The companion metrics we report (flip rate $(1 - \text{JSS})$, Cohen’s κ [3], and bootstrap 95% confidence intervals [5]) let readers separate raw agreement from chance-corrected agreement and quantify uncertainty.

Our contributions are: **(1) A systematic empirical study of prompt-induced decision instability** across four evaluation tasks (factuality, coherence, relevance, preference) and thirteen widely deployed judge models (seven commercial, six open-source). **(2) The JUDGESENSE benchmark.** We construct and release 500 hand-validated prompt-paraphrase pairs covering four distinct evaluation tasks, each with its own dedicated paraphrase set. Within each task, every pair consists of two instructions asking the judge to make the same evaluative decision on the same underlying item, with differences limited to surface wording. Items are drawn from established sources [7, 13, 20, 23]. All 500 pairs are hand-validated by a primary annotator and independently re-reviewed by a second annotator, with full agreement on the published set. **(3) Concrete empirical findings.** Coherence is the only

task that meaningfully separates judges; factuality is consistently high once polarity-inverted templates are removed; pairwise tasks reveal that twelve of thirteen judges anchor on option position rather than respond to prompt phrasing. Model size and architecture do not predict consistency: within-vendor inversions (Claude Opus 4.7 below Claude Haiku 4.5; GPT-5.5 below GPT-4o) and reasoning-tuned models clustering in the lower half of the coherence ranking show that frontier scale and recency are not reliable proxies for stable judging. **(4) Practitioner recommendations.** We map the findings to concrete guidance: which judges to use on which tasks, when to randomize option order, when prompt phrasing must be frozen, and what minimum stability bar (and reporting convention) production pipelines should adopt.

The rest of the paper is organized as follows. Section 2 situates JSS in the LLM-as-a-judge literature and distinguishes it from adjacent prompt-sensitivity work. Section 3 formalizes the framework: the JSS definition, dataset construction, paraphrase validation procedure, and statistical analysis. Section F pins down model checkpoints, inference settings, and reproducibility details. Section 4 reports the main results across all thirteen judges. Section E interprets the findings, gives practitioner recommendations, and notes limitations. Section 5 closes with the broader implications of the study and a call for paraphrase-stability reporting in future LLM-as-a-judge work.

2 Related Work

LLM-as-a-Judge. The use of strong language models as automated evaluators was popularized by Zheng et al. [23], who showed that GPT-4 agrees with crowdsourced human preferences on MT-Bench at a rate comparable to inter-annotator agreement among humans, and used this to justify replacing human raters at scale. Liu et al. [14] introduced G-Eval, a chain-of-thought prompting strategy that improves correlation with human ratings on summarization tasks. Chiang and Lee [2] examined whether LLM evaluations can substitute for human evaluations across a range of NLG tasks and found broadly positive but task-dependent agreement. These early results (high correlation on average, with caveats) drove adoption far ahead of methodological scrutiny. Two recent surveys [11, 12] document a literature in which LLM judges have become standard evaluation infrastructure across summarization, instruction following, retrieval, and reasoning benchmarks.

Bias and reliability of LLM judges. A second wave of work has documented systematic failure modes. Wang et al. [21] show that LLM judges exhibit position bias and verbosity bias on pairwise comparison tasks, with GPT-4 favoring whichever response appears first in the prompt. Shi et al. [18] extend this analysis to 12 judge models across more than 100,000 evaluation instances on MT-Bench and DevBench, and confirm that position bias is not a sampling artifact but a stable, model-specific property modulated by the quality gap between candidates. Thakur et al. [19] evaluate 13 judges on a clean alignment task and document sensitivity to prompt complexity, length, and other surface features alongside a general tendency toward leniency. These papers establish that judges are biased; what they do not provide is a reportable single-number metric for any given new judge model.

Prompt sensitivity of LLMs. A third line of work measures prompt sensitivity in LLMs more broadly. Elazar et al. [6] introduce a paraphrase-based consistency framework for pretrained language models, measuring whether equivalently phrased queries elicit the same factual completion; the agreement-rate-under-paraphrase quantity we adopt is mathematically of this same family. Sclar et al. [17] show that simple changes in formatting (delimiters, separators, whitespace) cause performance swings of 76 accuracy points on some benchmarks. Mizrahi et al. [15] argue for multi-prompt evaluation as a methodological default, on the grounds that any single-prompt result is a sample from a wide distribution. Zhuo et al. [24] introduce ProSA, which assesses prompt sensitivity in task performance across multiple LLM families. Razavi et al. [16] introduce PromptSET, a dataset for predicting prompt sensitivity on TriviaQA and HotpotQA, and find that current methods struggle to anticipate which prompts will trip a model up. All of this work measures the sensitivity of LLMs as *task-solvers*: how does varying the prompt change the answer that the LLM tries to produce? Our work measures the sensitivity of LLMs as *judges*: how does varying the prompt change the label the LLM assigns to someone else’s answer? The two are related but not identical, and the judge case has measurable implications for benchmarking practice that the task-solver case does not.

3 The JudgeSense Framework

3.1 Problem setup

Let j be a judge: a function that takes an evaluation prompt p as input and returns a decision d from a finite, task-specific decision space D_t . For factuality, $D_t = \{\text{YES}, \text{NO}\}$. For coherence, $D_t = \{1, 2, 3, 4, 5\}$ on a Likert scale. For relevance and preference, $D_t = \{\text{A}, \text{B}\}$. We are interested in the stability of j ’s decisions under paraphrase. A paraphrase pair is an ordered pair (p, p') such that a competent reader would judge p and p' to ask for the same evaluation, differing only in surface phrasing. A paraphrase set P is a collection of such pairs.

The companion metric to Equation 1 is the flip rate, $1 - \text{JSS}$. We also report Cohen’s κ [3] computed across all paraphrase pairs in a (model, task) cell, where the two “raters” are the same judge under prompt p and prompt p' . We report bootstrap 95% confidence intervals on JSS using $n = 1000$ resamples [5]. Two technical notes on κ are worth flagging up front. First, on coherence the Likert scale gives κ a meaningful chance-correction interpretation; on binary tasks κ is essentially a rescaled JSS. Second, when both prompt-variant decision streams collapse to a single label (the always-A behavior we discuss in Section 4), κ is mathematically undefined (0/0). Our implementation returns 1.0 in this degenerate case for tabulation purposes, but the value is not meaningful and we flag it accordingly throughout.

Measurement contract. JSS measures *decision consistency under paraphrase*; it does not measure accuracy, correctness, or alignment with human raters. This distinction matters in three ways. JSS is informative when (a) the judge exhibits non-degenerate behavior, using more than one label in D_t across the paraphrase set; (b) the paraphrase pairs are semantically validated, so that differences in decisions reflect wording sensitivity rather than genuine semantic differences; and (c) the decision semantics are stable across templates, meaning all

templates agree on what each label in D_t denotes. JSS is *misleading* in at least three failure modes. First, degenerate always-A behavior produces $JSS = 1.0$, a vacuously high score that carries no information about paraphrase sensitivity. Second, polarity-inverting templates (e.g., template 4 in our factuality suite, where YES means errors rather than correctness) make the judge appear inconsistent when the actual inconsistency is the evaluator’s label-mapping, not the judge’s decision process. Third, position-biased judges on binary tasks that anchor on one option under every phrasing achieve high JSS by a mechanism unrelated to instruction comprehension. Readers should interpret JSS alongside the degenerate-behavior flags and polarity-correction analyses reported in Sections 4 and 4.3.

Maintained assumptions. Three assumptions are required for JSS to be interpretable as a paraphrase-sensitivity estimate; we state them explicitly because each is violated in at least one part of our empirical study.

(A1) Paraphrase equivalence. Each pair (p_i, p'_i) differs only in surface phrasing, not in the semantic intent of the evaluation request. We validate this by hand review (primary annotator with independent re-review by a second annotator, criteria in Appendix B); the gpt-4o-mini classifier served as a first-pass screen whose disagreements with the human labels are reported in Section 3.3. Polarity-inverted template pairs were excluded from the published dataset after this review identified them as a systematic confound rather than a genuine test of paraphrase sensitivity.

(A2) Decision-space stability. The decision space D_t is interpreted uniformly across all template variants. This assumption is violated by polarity-inverting templates such as template 4 on factuality (full template list in Section 3.4), which formally keeps $D_t = \{\text{YES}, \text{NO}\}$ but inverts the label semantics (YES means errors rather than correctness). Without polarity correction, JSS conflates label-mapping disagreement with genuine paraphrase-induced inconsistency; corrected values are reported in Section 4.3.

(A3) Non-degenerate response. The judge uses more than one element of D_t across the paraphrase set. When a judge outputs the same label on every pair under every template (the always-A pattern on preference and relevance), JSS trivially equals 1.0; this value is not informative about sensitivity to prompt wording and should not be reported as evidence of consistency.

3.2 Dataset construction

We draw evaluation items from four established sources, one per task, to ensure that our prompts target tasks the community already cares about: **Factuality**, sourced from TruthfulQA [13]. Each item is a question paired with a candidate response; the judge decides whether the response is factually correct. **Coherence**, sourced from SummEval [7]. Each item is a summary; the judge rates coherence on a 1–5 Likert scale. **Relevance**, sourced from BEIR [20]. Each item is a query plus two candidate passages; the judge picks the more relevant. **Preference**, sourced from MT-Bench [23]. Each item is a query plus two candidate responses; the judge picks the preferred response.

For each task we authored five minimalist instruction templates that ask the same evaluative question in different surface phrasings (Section 3.4). All templates are instruction-only:

no chain-of-thought, no role-priming, no JSON envelopes. This isolates the effect of natural-language rewording. Within each task, four of the five templates are semantically equivalent; the fifth (Template 4 for factuality) deliberately inverts the decision polarity, making “YES” signal an incorrect response rather than a correct one. Template 4 pairings are excluded from the published dataset: during construction we found that polarity inversion introduces systematic label mismatch that conflates prompt-design error with judge inconsistency, making JSS on those pairs uninterpretable as a sensitivity measure. The per-template analysis and its impact on raw JSS are discussed in Section 4.3. For coherence, relevance, and preference, all five templates are semantically equivalent and 125 paraphrase pairs are generated per task. For factuality, pairs involving Template 4 are excluded, yielding 125 clean pairs. Total dataset: **500 pairs** across the four evaluation dimensions.

3.3 Paraphrase validation

Asserting that two prompts are paraphrases is itself a judgment that needs to be validated. The 500 published paraphrase pairs were hand-validated by the primary annotator and then independently re-reviewed by a second annotator (criteria documented in Appendix B). Each pair was labeled YES (semantically equivalent), NO (non-equivalent), or UNSURE (borderline). The gpt-4o-mini classifier served as a first-pass screen; we report its agreement with the human labels as a calibration check rather than as the primary validation.

Of the 500 published pairs, all 500 received YES labels from both annotators (0 NO, 0 UNSURE), giving full agreement between the two reviewers. The GPT-4o-mini classifier agreed with the annotator labels on 100% of all published pairs. Template 4 factuality pairings were identified during construction as polarity-inverted and excluded before publication; the 50 pairs involving Template 4 all received NO labels from both annotators (reversed label semantics constitute a genuine non-equivalence, not a surface paraphrase). The analysis of what JSS looks like when those pairs are included (and why including them produces the misleading roughly-37-percent flip rate observed in preliminary experiments) is discussed in Section 4.3.

3.4 Prompt template variants

Each task has five templates. The *factuality* templates are shown below as a worked example (binary YES/NO decision space, $D_t = \{\text{YES}, \text{NO}\}$). The Likert (1–5) coherence templates and the binary A/B relevance and preference templates follow the same minimalist structure with task-specific decision spaces; the complete set of 20 templates is provided in Appendix A. The templates are:

1. “Is this factually correct? Answer YES or NO only.”
2. “Fact-check this response. Reply YES (correct) or NO (incorrect).”
3. “Are the facts in this response accurate? YES or NO.”
4. “Does this response contain factual errors? Answer NO (accurate) or YES (has errors).”
5. “Evaluate factual accuracy. One word answer — YES or NO.”

Note that template 4 inverts the polarity of the answer relative to the others (NO means accurate, YES means errors). This template was excluded from the published dataset after human review identified it as a systematic confound: the reversed label convention is not merely a wording variation but a decision-space change that would always flip the apparent JSS regardless of model behavior. The per-template analysis showing the impact of this exclusion is in Section 4.3. Coherence, relevance, and preference templates follow the same minimalist pattern with task-appropriate decision spaces; the full set of 20 templates is in Appendix A.

3.5 Model selection

We selected thirteen judges spanning commercial and open-source families across a range of scales and instruction-tuning regimes. From the commercial side: GPT-5.5, GPT-4o, GPT-4o-mini (OpenAI); Claude Opus 4.7, Claude Sonnet 4.5, Claude Haiku 4.5 (Anthropic); and Gemini 2.5 Flash (Google). From the open-source side, accessed via hosted APIs: LLaMA-3.1-70B [9], Mistral-7B [10], Qwen-2.5-72B [22], DeepSeek-R1 [4], Qwen 3.6 Flash, and DeepSeek-V4 Flash. Five of these were chosen because they are the modal default judges in current benchmarking practice; the remaining eight expand the family and scale coverage to include frontier reasoning-tuned models that have become increasingly standard in the literature. Complete checkpoint identifiers, inference settings, and the evaluation protocol are in Appendix F and Appendix D.

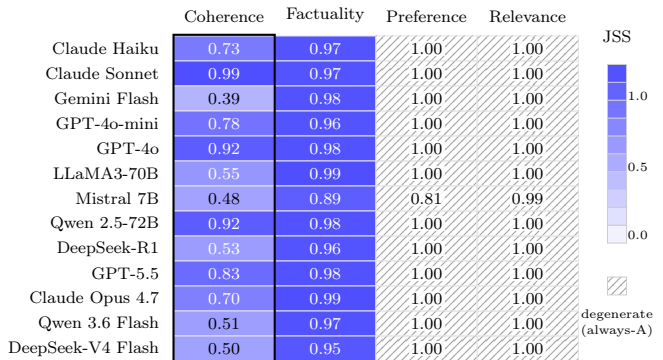


Figure 1: JSS across evaluation dimensions, with factuality shown after Template 4 polarity-inverted exclusion (see Section 4.3). Hatched cells indicate degenerate always-A behavior. Takeaway: only the coherence column carries judge-discriminating signal.

4 Results

Twelve of thirteen judges return label A on every preference and relevance pair regardless of prompt variant; $JSS = 1.000$ reflects position anchoring [18, 21], not consistency under paraphrase. We exclude these from the primary leaderboard and discuss them in Section 4.2.

4.1 Headline findings

Table 1 and Figure 1 report the core JSS results across all thirteen judges and four evaluation dimensions. We organize the discussion around five findings, ranked roughly by how surprising

Table 1: **JudgeSense main results.** JSS, flip rate, Cohen’s κ , and bootstrap 95% confidence interval for each (judge, task) cell. Rows are sorted by coherence JSS, descending. Coherence is the only task that meaningfully separates judges. Factuality JSS is consistently high across all judges (range [0.893, 0.987]; Template 4 polarity-inverted pairs excluded; see Section 4.3). Preference and relevance produce JSS = 1.000 for twelve of thirteen models under the original fixed-order design; this reflects position anchoring rather than genuine consistency. N is the post-exclusion sample size per cell. Takeaway: coherence is the only discriminating task; factuality and pairwise tasks are non-discriminating under the current benchmark design.

Model	Task	JSS	Flip	κ	95% CI	N
claude-sonnet-4-5	coherence	0.992	0.008	0.986	[0.981, 1.000]	375
qwen-2.5-72b	coherence	0.917	0.083	0.842	[0.888, 0.944]	375
gpt-4o	coherence	0.915	0.085	0.828	[0.888, 0.941]	375
gpt-5.5	coherence	0.827	0.173	0.694	[0.789, 0.864]	375
gpt-4o-mini	coherence	0.784	0.216	0.627	[0.744, 0.824]	375
claude-haiku-4-5	coherence	0.731	0.269	0.583	[0.688, 0.776]	375
claude-opus-4-7	coherence	0.701	0.299	0.580	[0.659, 0.747]	375
llama-3.1-70b	coherence	0.554	0.446	0.338	[0.488, 0.615]	260
deepseek-r1	coherence	0.528	0.472	0.332	[0.480, 0.576]	375
qwen-3.6-flash	coherence	0.512	0.488	0.372	[0.421, 0.560]	375
deepseek-v4-flash	coherence	0.495	0.505	0.349	[0.447, 0.543]	374
mistral-7b	coherence	0.480	0.520	−0.082	[0.429, 0.536]	375
gemini-2.5-flash	coherence	0.387	0.613	−0.057	[0.339, 0.435]	375
claude-opus-4-7	factuality	0.987	0.013	0.973	[0.973, 0.997]	375
llama-3.1-70b	factuality	0.986	0.014	0.971	[0.972, 0.997]	286
gpt-5.5	factuality	0.981	0.019	0.960	[0.968, 0.992]	373
qwen-2.5-72b	factuality	0.976	0.024	0.952	[0.957, 0.989]	375
gpt-4o	factuality	0.976	0.024	0.951	[0.960, 0.992]	375
gemini-2.5-flash	factuality	0.976	0.024	0.951	[0.959, 0.989]	369
qwen-3.6-flash	factuality	0.971	0.029	0.939	[0.952, 0.987]	374
claude-haiku-4-5	factuality	0.967	0.033	0.934	[0.948, 0.984]	366
claude-sonnet-4-5	factuality	0.965	0.035	0.930	[0.947, 0.981]	374
gpt-4o-mini	factuality	0.957	0.043	0.913	[0.936, 0.976]	375
deepseek-r1	factuality	0.957	0.043	0.913	[0.936, 0.976]	374
deepseek-v4-flash	factuality	0.947	0.053	0.890	[0.923, 0.965]	375
mistral-7b	factuality	0.893	0.107	0.776	[0.861, 0.925]	375

and actionable each is for practitioners choosing a judge.

Finding 1: Coherence is the discriminating task. Coherence JSS spans 0.387 (gemini-2.5-flash) to 0.992 (claude-sonnet-4-5), a gap of 0.605 across thirteen models (Figure 2). No other task in our suite shows comparable spread. On factuality, JSS is consistently high but narrowly spread across all judges, ranging from 0.893 to 0.987 (a gap of 0.094, compared to 0.605 for coherence); on preference and relevance, twelve of thirteen judges produce JSS = 1.000 for reasons we discuss below. We note that coherence is also the only task in

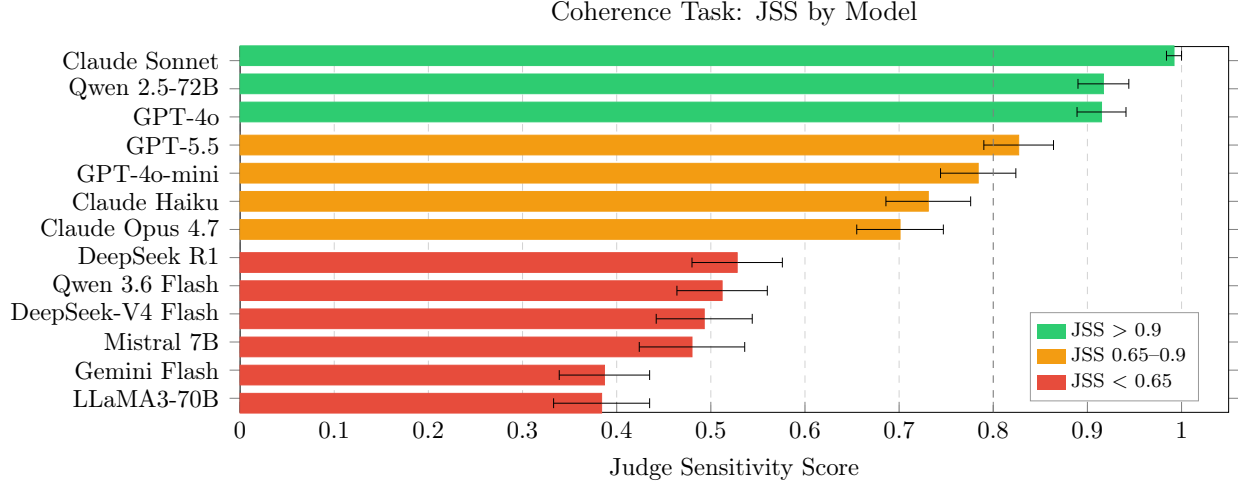


Figure 2: Coherence JSS by judge model. Error bars show 95% bootstrap confidence intervals. Takeaway: coherence JSS varies by more than 0.6 across thirteen judges, and the spread does not track model scale or recency.

our suite with a multi-class (1–5 Likert) decision space; the other three tasks are binary or ternary. The discriminating power of coherence is therefore confounded with label cardinality, and a controlled comparison that binarizes the Likert scale would be needed to fully separate the two effects (see Section E). If a practitioner wants to know whether their candidate judge is sensitive to prompt wording, coherence is where that question gets answered in the current benchmark. We recommend that future JSS reports lead with coherence and treat factuality, preference, and relevance as conditional add-ons. This recommendation holds within the current benchmark design; a second-iteration benchmark with A/B-randomized pairwise tasks may find that preference and relevance also carry discriminating signal.

Finding 2: Factuality JSS is consistently high across all judges (range [0.893, 0.987]). With polarity-inverted Template 4 pairings excluded from the published dataset, all thirteen judges produce factuality JSS in [0.893, 0.987], with most above 0.96. The high consistency across judges with different architectures, scales, and training regimes confirms that factuality, as operationalized by Templates 1–3 and 5, is weakly discriminating compared to coherence: all judges handle straightforward factual yes/no prompts with near-perfect stability. The preliminary experiments that retained Template 4 pairings showed a uniform flip rate near 37 percent across all thirteen judges, a rate so uniform it was clearly a dataset artifact rather than judge behavior. That analysis, which motivated the exclusion of Template 4 from the benchmark, is documented in Section 4.3 as a cautionary note for benchmark designers. Factuality is not a discriminating task in the published benchmark; coherence remains the only task that separates judges.

Finding 3: Scale is not a reliable predictor of consistency in our data. The two best coherence judges in our thirteen-model set are claude-sonnet-4-5 (0.992) and qwen-2.5-72b (0.917). The worst is gemini-2.5-flash (0.387), a model from Google’s frontier line. Within the

Anthropic family, claude-opus-4-7 (0.701) scores lower than claude-haiku-4-5 (0.731), which in turn scores far lower than claude-sonnet-4-5 (0.992), giving three models from the same provider that span 0.291 JSS units with the largest model ranking last. The Opus–Haiku gap is small in point-estimate terms and the bootstrap 95% confidence intervals overlap ([0.659, 0.747] vs. [0.688, 0.776]); the inversion is therefore directional, not statistically separated. GPT-5.5 (0.827) falls below GPT-4o (0.915), and the corresponding confidence intervals do not overlap ([0.789, 0.864] vs. [0.888, 0.941]), indicating a statistically distinct ranking inversion within the OpenAI family. qwen-3.6-flash (0.512) is far below qwen-2.5-72b (0.917) despite being a more recent release from the same family. DeepSeek-R1, run at `max_tokens=1024` in this study, falls to 0.528, a result we discuss in the Limitations section. In our thirteen-judge sample, raw parameter count and model recency do not predict coherence JSS. Plausible candidates include RLHF training strategy, the diversity of evaluation-style prompts in instruction tuning, and the degree to which the model has been explicitly trained to follow short constrained instructions. Our data do not allow us to isolate which of these factors matters; what the results establish is that selecting judges by parameter count or release date is an unreliable heuristic. Practitioners should measure JSS directly rather than infer it. We caution that this is an observational finding from a single benchmark; uncontrolled confounds including training data composition, instruction-tuning recipe, and inference configuration prevent causal attribution to model scale alone.

Finding 4: Mistral-7b is the only judge with non-degenerate preference behavior.

Twelve of the thirteen judges return the same preference label on every pair regardless of which template was used, producing $JSS = 1.000$ for those twelve. Under the balanced option-order design (62 of 125 preference pairs have the two responses swapped so the preferred response falls in position B), this uniform behavior reflects position anchoring rather than genuine paraphrase stability. The same pattern holds on relevance for twelve judges. The exception on preference is mistral-7b: $JSS = 0.811$, $\kappa = 0.542$, $N = 375$, under the balanced design. Mistral flips its preference verdict on 18.9% of paraphrase pairs, the only judge to show genuine paraphrase sensitivity on pairwise tasks. On relevance, mistral-7b produces $JSS = 0.989$ ($N = 375$), close to the degenerate ceiling but not reaching it. Two interpretations are consistent with this pattern. Either mistral-7b has weaker position-anchoring bias than the other twelve judges, in which case its non-degenerate preference result is the more honest consistency estimate; or mistral-7b is more susceptible to prompt phrasing on preference tasks while remaining consistent on relevance. We cannot fully distinguish these interpretations from the current data.

Finding 5: Negative kappa indicates systematic, not random, disagreement.

Two judges produce negative Cohen’s κ on coherence: gemini-2.5-flash ($\kappa = -0.053$) and mistral-7b ($\kappa = -0.082$). A negative κ means the two prompt variants disagreed *more often than chance*. Negative κ of this magnitude is unlikely to be sampling noise; rather, the two phrasings appear to shift the model toward consistently different regions of the Likert scale. For a judge to be useful, κ should at least be reliably positive; on coherence, eleven of thirteen judges clear that bar. Negative- κ judges are not merely inconsistent; their decisions under the two prompt variants are anti-correlated, and any pipeline that varies the evaluation prompt

across contexts should treat such judges as unreliable measurement instruments.

Finding 6: Extending the reasoning-trace token budget did not improve consistency on the one judge for which we have a before/after measurement. DeepSeek-R1 was run twice on coherence under different token budgets: once at `max_tokens = 20` (consistent with the standard-inference judges in pass 1) and once at `max_tokens = 1024` (consistent with the other reasoning-tuned judges). Coherence JSS *decreased* from 0.653 to 0.528 when the budget was extended, despite both runs being on identical paraphrase pairs. Allowing the full chain-of-thought to complete therefore did not stabilize the judge’s decisions and may have increased paraphrase-induced variance, possibly by giving the trace more surface area for divergence between the two paraphrased prompts. We flag this as a single-judge observation ($n = 1$ for the before/after comparison) rather than a generalizable claim; the four other reasoning-tuned judges (GPT-5.5, Claude Opus 4.7, Qwen 3.6 Flash, DeepSeek-V4 Flash) were measured only at `max_tokens = 1024`, so an analogous comparison is not available for them. The implication is methodological: when introducing reasoning-tuned judges into a paraphrase-stability benchmark, the token budget should be treated as a controlled variable, not assumed irrelevant. We discuss this further as a limitation in Section E.

4.2 Data quality and transparency

We report exclusion rates and degenerate-behavior caveats explicitly so that readers can re-derive any cell in Table 1 from the released decision logs.

API errors and malformed outputs. Vendor API errors were 0% for all judges except qwen-2.5-72b, which experienced 6.4% Novita timeouts. Malformed-output (UNCLEAR) rates were: 0% for claude-haiku, claude-sonnet, gpt-4o, deepseek-r1, gpt-5.5, claude-opus-4-7, and qwen-3.6-flash; under 1% for gemini-2.5-flash; 5.0% for gpt-4o-mini; 6.4% for qwen-2.5-72b (timeouts); 7.5% for llama-3.1-70b; and 15.7% for mistral-7b. For deepseek-v4-flash, UNCLEAR rates were 5.9% on preference and 9.3% on relevance; all other tasks were 0%. The mistral failures were almost entirely on coherence: rather than emitting a digit on the 1–5 Likert scale, mistral often produced a short English phrase (“somewhat coherent”, “the summary is coherent”), which our regex parser correctly flagged as malformed.

Safety refusals. There were zero safety refusals across all approximately 38,700 calls and all thirteen judges. None of the prompts in the dataset trigger safety filters at any of the providers we tested.

Always-A degenerate behavior. The original sweep used fixed option order; under that design, twelve of thirteen judges output “A” on 100% of preference and relevance decisions across both prompt variants, leaving two non-exclusive explanations indistinguishable: position bias [18, 21] or a source-benchmark artifact in which option A is genuinely preferred more often than chance. To separate these, we updated the dataset to a balanced A/B design (62 of 125 preference pairs and 63 of 125 relevance pairs have option order swapped, flagged via `ab_swapped`). Under the balanced design, eleven of thirteen judges still produce JSS = 1.000 with both decision streams collapsed to “A”, confirming that the always-A pattern is position anchoring rather than a fixed-order artifact. Mistral-7b is the only judge whose preference behavior becomes non-degenerate (JSS = 0.811, $N = 375$) under balanced ordering.

Excluded pairs. All Template 4 factuality pairings are excluded from the published dataset and from all metrics in Table 1. Factuality N is therefore $125 \times 3 = 375$ per cell; the exact N varies slightly by judge due to malformed-output exclusions.

4.3 Template 4 polarity inversion: a methodological note

Template 4 for the factuality task (“Does this response contain factual errors? Answer NO (accurate) or YES (has errors)”) inverts the decision polarity relative to Templates 1–3 and 5. Preliminary experiments that included Template 4 pairings in the factuality set produced a uniform flip rate near 37 percent across all thirteen judges, a rate so model-independent that it clearly reflects label-convention mismatch rather than judge behavior. This finding motivated the exclusion of Template 4 pairings from the published dataset. The key takeaway: polarity-inverting templates should be either excluded from paraphrase benchmarks or handled with explicit label remapping; failing to do so inflates apparent flip rates by ~ 37 percentage points uniformly across all judges.

4.4 Practitioner recommendations

We translate the empirical findings into per-task guidance.

High ($\text{JSS} \geq 0.90$, $\kappa \geq 0.80$): Safe for deployment in prompt-varying production pipelines. Judges in this tier (claude-sonnet-4-5 at 0.992, qwen-2.5-72b at 0.917, and gpt-4o at 0.915 on coherence) maintain consistent decisions even when different teams or evaluation rounds phrase the instruction differently.

Moderate ($0.80 \leq \text{JSS} < 0.90$): Acceptable for fixed-template, single-prompt pipelines; not recommended where prompt diversity is expected. Only one judge in our thirteen-model set falls in this range on coherence: gpt-5.5 (0.827). The coherence distribution clusters at the high end (three judges above 0.90) and at the low end (nine of the thirteen judges below 0.80), with a single model bridging the gap; the 0.80 threshold continues to mark a meaningful transition from acceptable to unreliable consistency on this task.

Low ($\text{JSS} < 0.80$): Treat the judge as a noise source for any application involving prompt variation. In our study this includes gpt-4o-mini (0.784), claude-haiku-4-5 (0.731), claude-opus-4-7 (0.701), llama-3.1-70b (0.554), deepseek-r1 (0.528), qwen-3.6-flash (0.512), deepseek-v4-flash (0.495), mistral-7b (0.480), and gemini-2.5-flash (0.387) on coherence. Two of these (mistral-7b and gemini-2.5-flash) have negative κ , meaning their two prompt-variant decision streams are anti-correlated.

Extended discussion, theoretical implications, broader impacts, and limitations are in Appendix E.

5 Conclusion

We introduced JUDGESENSE, a benchmark of 500 hand-validated prompt-paraphrase pairs across four tasks and thirteen judges. Our results show that coherence is the most discriminating task in our suite (JSS 0.387–0.992), that model scale and recency are not reliable proxies for consistency in our thirteen-judge sample, and that pairwise tasks are largely dominated

by position anchoring under the current benchmark design. We recommend measuring JSS directly rather than inferring it from parameter count, and suggest that pipelines relying on judges with JSS below 0.8 on the relevant task treat those results with additional caution.

Data and Code Availability

- **Code (MIT license):** <https://github.com/rohithreddybc/judgeSense>
- **Validated paraphrase dataset (CC-BY-4.0 license):** <https://huggingface.co/datasets/Rohithreddybc/judgesense-benchmark>
- **Per-call decision logs:** Included in the GitHub repository under `outputs/`.

Acknowledgements

This research used free-tier inference credits provided by Novita AI, HuggingFace, and Alibaba Cloud Model Studio (DashScope). No external funding was received.

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A Complete prompt template inventory

The full set of 20 prompt templates used in this study (5 per task) is shown below. All templates are minimalist instruction-only phrasings; no chain-of-thought, role-priming, or JSON envelopes are used.

Factuality (binary YES/NO).

1. “Is this factually correct? Answer YES or NO only.”
2. “Fact-check this response. Reply YES (correct) or NO (incorrect).”
3. “Are the facts in this response accurate? YES or NO.”
4. “Does this response contain factual errors? Answer NO (accurate) or YES (has errors).”
5. “Evaluate factual accuracy. One word answer — YES or NO.”

Coherence (Likert 1–5).

1. “Rate the coherence of this summary from 1 to 5.”
2. “On a scale from 1 to 5, how coherent is this summary?”
3. “Score this summary’s coherence (1 = incoherent, 5 = highly coherent). Reply with the digit only.”
4. “How well does this summary hang together? Rate 1–5.”
5. “Coherence rating for this summary, 1 to 5. One number only.”

Relevance (binary A/B).

1. “Which passage is more relevant to the query? Answer A or B.”
2. “Pick the more relevant passage for this query: A or B.”
3. “Given the query, which passage better matches the information need — A or B?”
4. “Compare the two passages against the query. Answer A or B.”
5. “Relevance judgment: A or B. One letter only.”

Preference (binary A/B).

1. “Which response is better? Answer A or B.”
2. “Choose the preferred response: A or B.”
3. “Given the user query, which assistant response is preferable — A or B?”
4. “Compare the two responses. Pick A or B.”
5. “Preference: A or B. One letter only.”

B Paraphrase validation criteria

All 500 published prompt-paraphrase pairs were reviewed by the primary annotator and then independently re-reviewed by a second annotator. Each pair (T_a, T_b) consists of two judge-prompt templates applied to the same underlying item. Each annotator assigned one of three labels:

- **YES** (semantically equivalent): both templates request the same evaluative judgment from the judge. A judge that answers correctly under one template would answer correctly under the other, modulo polarity remapping where applicable.

- **NO** (non-equivalent): the templates differ in what they ask the judge to evaluate, or differ in polarity without a straightforward remapping (e.g., Template 4 factuality pairs, where YES and NO carry reversed semantics).
- **UNSURE**: the boundary is genuinely ambiguous. Both annotators were instructed to use this label conservatively; it was never assigned.

Polarity-inverted pairs. Template 4 for the factuality task phrases the prompt as “does this response contain factual errors?”, making YES signal *inaccurate* rather than *accurate*. Pairs that combine Template 4 with any of Templates 1–3 or 5 were labeled NO by both annotators, because the reversed label convention constitutes a genuine semantic difference that could mislead a naive judge. These 50 pairs were excluded from the published dataset before release. The per-template JSS analysis showing the impact of including versus excluding polarity-inverted pairings is discussed in Section 4.3.

Outcome. Of the 500 published pairs reviewed: 500 labeled YES, 0 labeled NO, 0 UNSURE. The 50 Template-4 factuality pairings were excluded before publication; in pre-publication review they all received NO labels (reversed polarity constitutes genuine non-equivalence). Their exclusion is the design decision that yields a clean 500-pair benchmark.

C Bootstrap procedure

Confidence intervals on JSS are computed as follows. For a given (judge, task) cell with N paraphrase pairs and observed per-pair agreements $a_1, a_2, \dots, a_N \in \{0, 1\}$, we draw $n = 1000$ resamples of size N with replacement from $\{a_i\}$. For each resample b we compute $\widehat{\text{JSS}}_b = \frac{1}{N} \sum_i a_i^{(b)}$. The reported 95% confidence interval is the empirical [2.5, 97.5] percentile range over the $n = 1000$ resamples. The procedure follows Efron [5].

D Evaluation Protocol

For each (judge, task, paraphrase pair) tuple we collect a single decision per template, with greedy decoding (`temperature` = 0.0 where supported, 0.01 on HuggingFace endpoints) and a per-class token budget. Non-reasoning instruction-tuned judges (`max_tokens` = 20) commit to a label within the first output tokens regardless of budget; a 50-pair spot-check on GPT-4o and LLaMA-3.1-70B at `max_tokens` = 1024 produced identical decisions, confirming that the short budget does not truncate informative content for this class. Reasoning-tuned judges (DeepSeek-R1, GPT-5.5, Claude Opus 4.7, Qwen 3.6 Flash, DeepSeek-V4 Flash) are given `max_tokens` = 1024 so their chain-of-thought can complete before the answer token is emitted. This creates a necessary but acknowledged comparison asymmetry: reasoning-tuned and standard-inference judges operate under different token budgets, so differences in their JSS values cannot be unambiguously attributed to model architecture versus inference configuration (see Limitations, Section E). Outputs are normalized via regex string matching against the task-specific decision space; decisions that fail to parse are logged

as UNCLEAR and excluded from JSS calculation, with the exclusion rate reported per model. We perform three runs per pair and report decisions from the first successful pass; the planned $500 \times 2 \times 3 \times 13 = 39,000$ judge calls completed at approximately 38,700 due to provider quota limits and rate-limit retries, giving an effective coverage of 99.2%.

E Discussion

E.1 What the coherence rankings actually mean

The coherence column of Table 1 is the most discriminating dimension across our thirteen judges. Two related questions follow. Why does coherence behave this way and the other tasks do not? And what should a practitioner do with the rankings?

On the first question, coherence is the only task in our suite with a non-binary decision space (1–5 Likert). The Likert scale gives the judge five places to land instead of two, which makes paraphrase-induced shifts visible: a judge that drifts from “4” under prompt p to “3” under prompt p' is registered as a flip, whereas a judge that drifts from “YES” under p toward “YES” under p' but with lower latent probability is registered as consistent. Said differently, binary tasks compress the consistency signal; Likert tasks expose it. This suggests that future work on judge sensitivity should preferentially use scaled, multi-class decision spaces if the goal is to characterize the judge rather than to certify a particular pipeline. Mizrahi et al. [15] make a related argument for multi-prompt evaluation more broadly: single-prompt results are samples from a wide distribution, and the variance is what we should be measuring.

Why might coherence differentiate judges while other tasks do not? Three properties make coherence the discriminating dimension in our suite. First, the five-point Likert scale exposes sub-threshold shifts that a binary decision space compresses away: a judge drifting from rating 4 to rating 3 is registered as a flip, whereas the equivalent drift on a binary YES/NO item is invisible. Second, coherence requires holistic text quality assessment rather than fact retrieval or forced pairwise ranking, leaving more room for prompt phrasing to shift the judge’s interpretive frame. Third, coherence is the most subjective of the four tasks; Likert-scale instructions that differ in wording (“rate” vs. “score” vs. “how coherent”) may shift the implicit anchor differently across models with different instruction-tuning mixtures. Among model-level factors, instruction-following discipline, RLHF training on evaluation-style prompts, and Likert-scale exposure during pretraining are all plausible contributors to the spread. We cannot isolate these from our results and flag this as a priority for future ablation work.

On the second question, the rankings are usable as-is. Claude-sonnet-4-5, qwen-2.5-72b, and gpt-4o all produce coherence JSS above 0.9, with κ above 0.8 (chance-corrected agreement in the “substantial” range). These are reasonable defaults for production judging pipelines that vary prompt phrasing across contexts. Gpt-5.5 (0.827) occupies a moderate tier, with JSS in the 0.80–0.90 range; this level of consistency is acceptable for fixed-template settings but may be insufficient where prompt diversity is expected. The next tier encompasses gpt-4o-mini (0.784) and claude-haiku-4-5 (0.731); 20–35% of decisions may flip under rephrasing. The lower tier includes claude-opus-4-7 (0.701), llama-3.1-70b (0.554), deepseek-r1 (0.528),

qwen-3.6-flash (0.512), deepseek-v4-flash (0.495), mistral-7b (0.480), and gemini-2.5-flash (0.387); these should not be used as coherence judges where prompt phrasing varies. Two of these (mistral-7b and gemini-2.5-flash) have negative κ , which means their decisions are anti-correlated with themselves under paraphrase.

E.2 Theoretical implications

Our systematic study illuminates one slice of the LLM-as-a-judge reliability question. Prior work measures a judge’s agreement with human raters or with a gold label. The JudgeSense benchmark instead enables measuring the judge’s agreement *with itself* under semantically equivalent prompts. These two notions of reliability are independent. A judge can have high human agreement but low JSS (picks the right answer, but which template determines the pick). A judge can also have high JSS but low human agreement (consistent but wrong). Paraphrase-based self-consistency measurement complements human-agreement measures; both should be reported.

A natural follow-up is a structural decomposition: how much of a judge’s decision-to-decision variance under paraphrase is attributable to template-level confounds (e.g., decision-polarity inversion) versus genuine wording-level sensitivity? Our paraphrase set was authored to minimize the former; Section 4.3 shows that polarity inversion constitutes a structural confound that can be eliminated by careful template selection before publication. Decomposition tools from the prompt-sensitivity literature [17, 24] could plausibly be adapted to this purpose, and we leave it to future work.

The benchmark in the broader evaluation landscape. The JudgeSense benchmark and the JSS measurement it enables are complementary to three existing approaches for managing prompt sensitivity, and it is useful to characterize what distinguishes this approach from each. Mizrahi et al. [15] advocate for multi-prompt evaluation as a methodological default: running evaluation across a distribution of prompts and averaging results to absorb sensitivity. This strategy reduces the impact of any single bad prompt, but it requires re-running the full evaluation with each new judge model on a study-specific prompt distribution; measuring against a fixed, validated paraphrase set like JudgeSense, by contrast, yields a judge-level property computed once and portable to any evaluation context. Variance-based approaches [17, 24] quantify the performance gap across prompt variants, measuring how much accuracy changes as the prompt changes; these methods require gold labels to compute accuracy at each prompt variant, restricting their applicability to settings where ground truth exists. The JSS measurement requires *no gold labels*: it measures self-consistency, comparing the judge’s decision under one phrasing against its own decision under another, making it applicable wherever a judge is used, even when ground truth is unavailable. Ensemble judging [11] manages sensitivity by combining multiple judges, averaging away individual judge variability; this strategy does not reveal whether any individual judge is stable enough to use alone, and it does not inform which judges should or should not be included in the ensemble. Measuring JSS against the JudgeSense benchmark characterizes each judge independently, making it directly applicable to both judge selection and ensemble design. The key property that this benchmark-based measurement offers across all three comparisons is portability and

gold-label-freedom: a judge’s self-consistency can be reported in a single row of a results table for any judge on any task where a validated paraphrase set exists.

E.3 Broader Impacts

JudgeSense benefits the research community by enabling quantitative, prompt-agnostic selection of LLM judges, reducing the risk of drawing incorrect conclusions from prompt-dependent automated evaluations. Practitioners gain a concrete tool for auditing, regression testing, and standardizing judge selection before deploying LLM-as-a-judge pipelines in high-stakes settings such as model development, scientific benchmarking, and content moderation.

A potential adverse use is *gaming*: a malicious actor who knows a judge’s JSS profile could craft prompts that exploit its known sensitivities to obtain a desired verdict in automated evaluation pipelines. We mitigate this risk by (a) publishing JSS results publicly so pipeline designers can proactively select robust judges with high JSS, (b) recommending A/B option-order randomization as a complementary defense against position-anchoring, and (c) framing JSS as a diagnostic rather than a certification, which discourages treating any single judge as infallible. The benchmark itself contains no sensitive personal data; it is derived entirely from publicly available datasets (TruthfulQA, SummEval, BEIR, MT-Bench) under permissive licenses, with no collection of new human data.

E.4 Limitations

Several limitations of this study are worth flagging explicitly.

Factuality task is non-discriminating. As discussed in Finding 2, factuality JSS is consistently high (range [0.893, 0.987]) across all thirteen judges in the published dataset. Polarity-inverted Template 4 pairings were excluded before publication precisely because they constituted a systematic confound (see Section 4.3). The factuality column of Table 1 confirms high consistency across all judges but should not be used to rank them; the task does not discriminate under the current published paraphrase set.

Position bias in pairwise tasks. Under the original fixed-option-order design, twelve of thirteen judges consistently selected option A on preference and relevance tasks, producing $\text{JSS} = 1.000$ values that are uninformative about paraphrase sensitivity. We updated the dataset to a balanced A/B design (62 of 125 preference pairs and 63 of 125 relevance pairs have option order swapped, flagged via `ab_swapped`), allowing position bias to be separated from paraphrase sensitivity. Under the balanced design, eleven of thirteen judges still produce $\text{JSS} = 1.000$ with both decision streams collapsed to label “A”, confirming that the always-A pattern reflects position anchoring rather than a fixed-order artifact; mistral-7b is the only exception on preference ($\text{JSS} = 0.811$).

Reasoning-tuned model token budget. Five of the thirteen judges are reasoning-tuned models that generate an internal chain-of-thought before committing to a final answer: DeepSeek-R1, GPT-5.5, Claude Opus 4.7, Qwen 3.6 Flash, and DeepSeek-V4 Flash. All five were run at `max_tokens` = 1024 to allow their reasoning traces to complete. DeepSeek-R1 is the only judge for which we have a before/after comparison: in the original pass at `max_tokens` = 20, it produced a coherence JSS of 0.653; at `max_tokens` = 1024, the JSS fell to 0.528. Completing the full reasoning trace therefore did not improve consistency and may

have increased paraphrase-induced variance. The four remaining reasoning-tuned models (GPT-5.5, Claude Opus 4.7, Qwen 3.6 Flash, DeepSeek-V4 Flash) were measured only at `max_tokens` = 1024, so no equivalent before/after comparison is available for them; their coherence JSS values are 0.827, 0.701, 0.512, and 0.495 respectively. Whether this positioning is attributable to the token budget, to architectural properties of reasoning-tuned models, or to both, remains an open question. We report the 1024-token measurements throughout and flag the token budget as a variable that warrants dedicated investigation in follow-up work. The practical implication is that the coherence JSS values for reasoning-tuned models should not be interpreted as evidence that reasoning capability reduces consistency; the token budget difference is an uncontrolled confound.

Limited annotator pool. Paraphrase equivalence was certified by the primary annotator and independently re-reviewed by a second annotator, who agreed on all 500 published pairs. A more robust design would engage three or more annotators on a deliberately contested edge-case subset and report Cohen’s κ across reviewers. The exclusion of polarity-inverted template pairs from the published dataset reflects a judgment confirmed by both reviewers; while the decision is well-motivated (see Appendix B), broader committee review of boundary cases on Template 4 inversion has not been performed.

English-only. All paraphrase pairs are in English. Multilingual JSS (does a judge consistent in English remain consistent under translation-equivalent prompts in other languages?) is an obvious extension, likely to surface different rankings.

Single-pass evaluation. We collect one decision per (judge, pair, template) tuple at temperature zero. Sampling-based JSS at higher temperatures would also be informative, particularly for understanding whether the consistency that claude-sonnet-4-5 shows at $T = 0$ survives at $T = 1.0$.

Single-vendor access per judge. Each judge is accessed through a single provider, so transient vendor-side changes (model alias drift, replica-level routing, or undocumented checkpoint updates) cannot be cross-validated against an independent host. We mitigate by pinning checkpoint identifiers where the vendor exposes them (Section F), but the `mistral-small-latest` alias and similar floating identifiers remain a residual confound that would require multi-host re-runs to fully eliminate.

F Experimental Setup

F.1 Model checkpoints

All judges were called via their respective vendor or hosted-inference APIs. We pin checkpoints where the vendor exposes a stable identifier:

- **gpt-4o**: `gpt-4o-2024-08-06` (OpenAI API).
- **gpt-4o-mini**: `gpt-4o-mini-2024-07-18` (OpenAI API).
- **claude-sonnet-4-5**: `claude-sonnet-4-5` (Anthropic API, accessed April 2026).
- **claude-haiku-4-5**: `claude-haiku-4-5-20251001` (Anthropic API).

- **gemini-2.5-flash**: Google Gemini API [8].
- **llama-3.1-70b**: meta-llama/Llama-3.1-70B-Instruct via HuggingFace Inference API [9].
- **mistral-7b**: mistral-small-latest via Mistral API [10]; the floating alias resolved to a single underlying checkpoint over the run window.
- **qwen-2.5-72b-instruct**: qwen/qwen-2.5-72b-instruct via Novita AI [22].
- **deepseek-r1**: deepseek/deepseek-r1 via Novita AI [4]. This is the reasoning-tuned R1 model, not v3.
- **gpt-5.5**: gpt-5.5 (OpenAI API, April 2026).
- **claude-opus-4-7**: claude-opus-4-7 (Anthropic API, April 2026).
- **qwen-3.6-flash**: qwen3.6-35b-a3b via Alibaba Cloud DashScope (April 2026).
- **deepseek-v4-flash**: deepseek/deepseek-v4-flash via Novita AI (April 2026).

F.2 Inference settings

We use greedy decoding (`temperature` = 0.0, except HuggingFace endpoints which require a strictly positive value; we set 0.01 there). `top_p` is left at the provider default. Non-reasoning instruction-tuned judges (`max_tokens` = 20) produce their label within the first tokens regardless of budget; reasoning-tuned judges (DeepSeek-R1, GPT-5.5, Claude Opus 4.7, Qwen 3.6 Flash, DeepSeek-V4 Flash) are given `max_tokens` = 1024 so their chain-of-thought can complete before the answer token. The system prompt is “*You are an evaluation assistant. Give only the requested answer with no explanation.*” for OpenAI and Anthropic; on HuggingFace, Novita, and DashScope endpoints the system prompt is delivered as a user-turn prefix because the underlying clients do not transmit a separate system role. For Gemini we explicitly disable the thinking budget (`thinking_budget` = 0) so that the model returns a direct answer rather than chain-of-thought tokens.

F.3 Inference stack

Calls were issued from a single Python process per judge, using vendor-supplied SDKs: `openai` for OpenAI models, `anthropic` for Claude, `google-generativeai` for Gemini, `mistralai` for Mistral, `huggingface_hub.InferenceClient` for the HuggingFace-hosted Llama-3.1-70B, a small custom HTTP client for Novita endpoints (DeepSeek-R1, DeepSeek-V4 Flash), and the OpenAI-compatible DashScope client for Qwen 3.6 Flash. No local inference was performed; all judges ran on vendor infrastructure.

F.4 Total calls, cost, and wall-clock

The full sweep called for $500 \text{ pairs} \times 3 \text{ runs} \times 2 \text{ prompts} \times 13 \text{ models} = 39,000$ judge invocations across all four tasks (factuality: 125 pairs; coherence, preference, relevance: 125 pairs each). Approximately 38,700 (99.2%) completed successfully; the shortfall is accounted for by provider rate-limit retries and intermittent Novita timeouts for deepseek-v4-flash on a small fraction of relevance and preference pairs. The nine-model pass-1 sweep (April 22–24, 2026) cost approximately \$11.55 USD; the pass-2 sweep adding four reasoning-tuned judges cost \$10.10 USD in total (OpenAI/GPT-5.5: \$3.36; Novita/DeepSeek-V4 Flash: \$3.67; Anthropic/Claude Opus 4.7: \$2.07; Alibaba Cloud DashScope/Qwen 3.6 Flash: \$1.00 after the provider’s 1M-token free quota). The combined cost for the full thirteen-judge evaluation was approximately \$21.65 USD. Wall-clock time for the full thirteen-judge evaluation was approximately five days running models sequentially with limited within-model parallelism. All inference was performed via vendor cloud APIs; no local GPU or CPU cluster was used.

F.5 Reproducibility

Code, paraphrase prompts, and the per-call decision logs are released at <https://github.com/rohithreddybc/judgeSense> under MIT license. The validated paraphrase dataset (500 pairs) is publicly available on HuggingFace under a CC-BY-4.0 license at <https://huggingface.co/datasets/Rohithreddybc/judgesense-benchmark>. All inference settings, system prompts, and parsing regular expressions are checked into the repository so that a third party can replicate any cell of the main results table given vendor API access.